METRIC

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DETAIL SPECIFICATION

PLASTIC MOLDING and EXTRUSION MATERIAL, POLYETHERETHERKETONE (PEEK)

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers polyetheretherketone (PEEK) thermoplastic materials suitable for injection molding, extrusion, or compression molding (See 6.1).

1.2 Classification. The PEEK thermoplastic materials should be of the following types and classes (see 6.2):

1.2.1 Type I. Type I PEEK thermoplastic materials are nonreinforced plastic.

1.2.2 Type II. Type II PEEK thermoplastic materials are Glass Fiber Reinforced Plastic (GFRP)

Class 1 at 10 ± 2% GFRP (by wt).

Class 2 at 20 ± 2% GFRP (by wt).

Class 3 at 30 ± 2% GFRP (by wt).

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.2.3 Type III. Type II PEEK thermoplastic materials are Carbon Fiber Reinforced Plastic (CFRP).

Class 1 at 20 ± 2% CFRP (by wt).

Class 2 at 30 ± 2% CFRP (by wt).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.3 Non-government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

ASTM INTERNATIONAL

ASTM D149	-	Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies
ASTM D256	-	Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics
ASTM D257	-	D-C Resistance or Conductance of Insulating Materials
ASTM D570	-	Water Absorption of Plastics
ASTM D638	-	Standard Test Method for Tensile Properties of Plastics
ASTM D648	-	Deflection Temperature of Plastics Under Flexural Load
ASTM D790	-	Flexural Properties of Unreinforced and Reinforced Plastics
ASTM D792	-	Specific Gravity and Density of Plastics by Displacement
ASTM D2863	-	Measuring the Minimum Oxygen Concentration to Support Candle Like Combustion of Plastics (Oxygen Index)
ASTM D3171	-	Fiber Content of Resin-Matrix Composites by Matrix Digestion
ASTM D3418	-	Transition Temperatures of Polymers by Thermal Analysis
ASTM E662	-	Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials

(Copies of these documents are available online at https://www.astm.org/).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes

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precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material shall consist of PEEK thermoplastic granules, pellets or powder, with or without fiber reinforcement.

3.2 Physical and mechanical properties. The properties of the PEEK thermoplastic shall be as specified herein when tested in accordance with 4.3. Mechanical properties shall be measured on specimens that are (a) injection molded to net shape, or (b) machined to the required specimen geometry from representative stock shapes produced by the applicable fabrication method (e.g., extruded or compression-molded plate/billet).

3.2.1 Nonreinforced PEEK (type I). The properties of the unpigmented, non- reinforced PEEK thermoplastic shall be as specified in table I.

TABLE I. Properties of nonreinforced PEEK (type I).

Property	Property Values
Melting point (minimum)	334°C (633°F)
Glass transition temperature (T_g), min	143°C (289°F)
Specific gravity (g/cm^3)	1.30 ± 0.04
Tensile strength [1/] (minimum) (at 23°C)	90 MPa (13,000 psi)
Tensile strength [1/] (minimum) (at 200°C)	20 MPa (2,900 psi)
Elongation at break (minimum)	10%
Flexural modulus (minimum)	3,400 MPa (500,000 psi)
Heat deflection temperature (minimum) (at 1,820 kPa)	149°C (300°F)
Water absorption (maximum) (24-h immersion)	0.2%

[1/] MPa x 145 = psi.

3.2.2 Glass fiber reinforced PEEK (type II). The properties of glass fiber reinforced PEEK thermoplastic shall be as specified in table II.

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TABLE II. Properties of glass fiber reinforced PEEK (type II).

Property	Density		
	Class 1 (l0% glass fiber)	**Class 2 (20% glass fiber)**	**Class 3 (30% glass fiber)**
Glass fiber content	10 ± 2% (by wt)	20 ± 2% (by wt)	30 ± 2% (by wt)
Specific gravity (g/cm^3)	1.37 ± 0.04	1.44 ± 0.04	1.49 ± 0.04
Tensile strength 1/ (minimum)	110 MPa (16,000 psi)	120 MPa (17,400 psi)	152 MPa (22,000 psi)
Elongation at break (minimum)	5%	4%	3%
Flexural modulus (minimum)5	5,000 MPa (730,000 psi)	6,200 MPa (900,000 psi)	6,900 MPa (1,000,000 psi)
Impact strength 2/ (minimum)	50 J/m (0.9 ft-lbf/in)	60 J/m (1.1 ft-lbf/in)	80 J/m (1.5 ft-lbf/in)

1/ The procedure in paragraph 4.5 shall be used to determine water absorption.
2/ Notched, 0.25 in; J/m x 18.73 x 10-3 = ft-lbf/in.

3.2.3 Carbon fiber reinforced PEEK (type III). The properties of carbon fiber reinforced PEEK thermoplastic shall be as specified in table III.

TABLE III. Properties of carbon fiber reinforced PEEK (type III).

Property	Density	
	Class 1 (20% carbon fiber)	**Class 2 (30% carbon fiber)**
Carbon fiber content	20 ± 2% (by wt)	30 ± 2% (by wt)
Specific gravity (g/cm^3)	1.38 + 0.04	1.42 + 0.04
Tensile strength 1/ (minimum)	159 MPa (23,000 psi)	185 MPa (27,000 psi)
Elongation at break (minimum)	5%	1%
Flexural modulus (minimum)5	10,300 MPa (1,500,000 psi)	12,800 MPa (1,850,000 psi)
Impact strength 2/ (minimum) (notched, 0.25 in)	45 J/m (0.8 ft-lbf/in)	50 J/m (0.9 ft-lbf/in)
Impact strength 2/ (minimum) (unnotched, 0.25 in)	400 J/m (8.0 ft-lbf/in)	530 J/m (10.0 ft-lbf/in)

3.3 Form. The form of the PEEK thermoplastic shall be granules, pellets or powder. When specified by the procuring activity, the form shall be in the size and shape specified (see 6.2).

3.4 Flammability and burning.

3.4.1 Oxygen index. The oxygen index for all types of PEEK thermoplastic shall be 35% minimum (see 4.3). The thickness of the test specimen shall be 3.2 mm.

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3.4.2 Specific optical density. The maximum specific optical density, D_{max}, in the flaming mode for all types of PEEK thermoplastic shall be 25 (see 4.3 and 6.3). The D_{max} in the nonflaming mode shall be 5. Test Samples for smoke emission tests shall be prepared as slabs 0.125 in thick x 3 in x 3 in (3.1 mm thick x 76.2 mm x 76.2 mm) in accordance with ASTM E662.

3.5 Electrical properties. The electrical properties of the PEEK thermoplastic shall be as specified in table IV (see 4.3).

TABLE IV. Electrical properties of PEEK.

Property	Minimum Property Values		
	Type 1 (nonreinforced)	Type 2 (30% glass fiber)	Type 3 (30% carbon fiber)
Volume resistivity 1/	10^{15} ohm-cm	10^{15} ohm-cm	1200 ohm-cm
Dielectric strength 2/	400 volts/mil (15.8 MV/m)	450 volts/mil (17.7 MV/m)	---

1/ Volume resistivity in ohm-m has a value of 1/100 of the value in ohm-cm.
2/ The thickness of the test specimen shall be 0.125 in (3.1 mm).

3.6 Ozone depleting substances. Class I and II ozone depleting substances (ODS) shall not be used in MIL-DTL-46146 or any referenced procedures.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The examination and testing of the plastic molding and extrusion material shall be classified as follows:

a. Preproduction tests
b. Quality conformance tests

4.2.1 Preproduction tests. Unless otherwise specified (see 6.2), preproduction tests shall consist of tests for the following properties:

a. Flexural modulus.
b. Heat deflection temperature.
c. Water absorption.
d Impact strength.
e. Oxygen index.

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f. Specific optical density.

g. Volume resistivity.

h. Dielectric strength.

The supplier shall furnish to the procuring activity (in lieu of continuous lot-to-lot testing) a certificate of compliance, which certifies that the requirements for the properties listed in (a) through (h) above have been met (see 3.2 through 3.5).

4.2.2 Quality conformance tests. Unless otherwise specified (see 6.2), quality conformance tests shall consist of tests for the following properties:

a. Melting point

b. Specific gravity

c. Tensile strength

d. Elongation at break

e. Fiber content

4.2.2.1 Lot size. Lot size shall consist of all the molding or extrusion material manufactured at one time, from one batch, forming part of one contract or order, and submitted for acceptance at the same time and place.

4.2.2.2 Sampling. Unless otherwise specified (see 6.2), a sufficient quantity of material from one unit representative of each lot (4.2.2.1) shall be selected at random for quality conformance testing. Each container of material shall be considered as a unit of product. Failure of sample to meet the quality conformance requirements shall be cause for lot rejection.

4.3 Test methods. The properties of the PEEK thermoplastic materials specified in 3.2 through 3.5 shall be determined in accordance with the applicable test method in table V.

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TABLE V. Test Methods.

Property	Test Method
Melting point (minimum)	ASTM D3418
Glass transition temperature (T_g)	ASTM D3418
Specific gravity	ASTM D792
Tensile strength	ASTM D638
Elongation at break	ASTM D638
Flexural modulus	ASTM D790
Heat deflection temperature	ASTM D648
Water absorption	ASTM D570
Fiber content (types II and III only)	ASTM D3171
Izod impact strength	ASTM D256
Oxygen index	ASTM D2863
Specific optical density	ASTM D814
Volume resistivity	ASTM D257
Dielectric strength	ASTM D149

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Polyetheretherketone (PEEK) thermoplastic materials are intended for use as a matrix for glass fiber and carbon-graphite composites, for injection and compression molded parts, monofilament, insulation for electrical wire and cable, and as feedstock for Additive Manufacturing (AM). The material has very low flammability, low smoke and toxic gas emission, and resistance to gamma radiation.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type and class of material.
c. Specific pellet size, if required (see 3.3).
d. Whether a preproduction sample is required and, if so, pertinent details (see 4.2.1).
e. Quality conformance tests, if other than specified in 4.2.2.
f. Sampling plan, if other than specified in 4.2.2.2.
g. Level of packing required (see 5.1).

6.3 Specific optical density. The maximum specific optical density, D_{max}, represents total smoke accumulation.

6.4 Gamma radiation resistance. Polyetheretherketone is extremely resistant to embrittlement by gamma radiation. Wire samples, coated with various grades (molecular weights) of PEEK polymer to a total diameter of 1.5 mm, have been exposed to gamma radiation from ^{60}Co (Cobalt-60) spent fuel cells. Samples were removed at 250 Mrad intervals until a maximum dose of 2000 Mrad had been absorbed. The samples were subjected to a wrap test around a 1D mandrel followed by a standard dielectric test (1.5 kV for 1 minute in water). In the range suggested for high performance wire covering (melt viscosity = 0.4 to 0.5 $kNsm^{-2}$), the dosage level of gamma radiation required to cause failure of the insulation is of the order of 1100 Mrad (11 MGy). Note: 1 rad = 0.01 gray.

6.5 PEEK varieties. Significant advancements in PEEK have been made since the prior revision to this specification. In addition to inclusion of types listed within this specification, additional types of PEEK are now available. Table VI lists descriptions and applications of commercially available PEEK varieties.

TABLE VI. Commercially available PEEK.

PEEK Type	Description
Virgin PEEK	High purity PEEK with good elongation
Glass-Filled PEEK (30%)	Enhanced stiffness and dimensional stability
Carbon-Filled PEEK (30%)	Improved strength, conductivity, and wear resistance
Bearing-Grade PEEK	Optimized for low friction and high wear applications
Medical-Grade PEEK	USP Class VI & ISO 10993 compliant for implantables
FDA-Approved PEEK	Suitable for food contact and sanitary processing
AM Feedstock PEEK	PEEK in powder or filament spools for AM applications
ESD-Safe PEEK	Electrically conductive or static-dissipative grades for electronics and semiconductor use

6.6 PEEK for AM. PEEK is commercially available in filament and powder forms as a feedstock material for AM (3D printing).

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6.7 Subject term (key word) listing.

3D Printing	Compression molding	Filament
Additive Manufacturing	Extrusions	Glass-Filled
Carbon-Filled	Feedstock	Injection molding

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Air Force - 20

Preparing activity:
ARMY - MR

(Project 9330-2025-005)

Review activities:
Army - AT, AV, GL, MD
Air Force - 03, 184

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